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Studies on the microstructures and properties in phase transformation of homogenized 7050 alloy

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ABSTRACT

The evolutions of mechanical properties and the morphologies of secondary phases in homogenized 7050 alloy were studied in detail by conductivity measurement, tensile test, scanning electron microscope (SEM), energy dispersive spectrometry (EDS), electron probe microanalysis (EPMA) and X-ray diffraction (XRD). Both σ_b and δ of the 7050 alloy increase with increasing homogenized temperature, and reach the saturation values at 440 °C/2 h. The secondary phases in as-cast 7050 alloy consist of T (Al_4Mg_2CuZn or $Al_3Mg_3CuZn_2$), S (Al_2CuMg) and Al_7Cu_2Fe. During homogenization, the region mainly situated at the margin of T phase having (wt%) 2.3–5.8 Mg, 72.2–82 Al, 5.4–14.8 Cu and 6.3–7.2 Zn sluggishly dissolves into the α(Al) matrix; at the same time, an elemental diffusion cell network appears within T phase, composed of the white cell wall having (wt%) 12.2–14 Mg, 44.9–51.6 Al, 20.4–27.4 Cu and 11.4–15.8 Zn and the light gray cell interior (wt%) 13.8–15.2 Mg, 43.9–48.1 Al, 26–33.3 Cu and 5.6–10 Zn. The network fades gradually when the light gray cell interior transforms to the gray phase and then to the dark gray phase, and disappears finally when the composition of the equilibrium S phase approaches. The diffusion of Zn happens even at 380 °C, and while the obvious diffusions of Mg and Cu begin at 440 °C. S phase contains 77% of the residual phases in 7050 alloy homogenized at 450 °C/2 h.

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1. Introduction

7050 Alloy has a combination of high strength and fracture toughness, low density and good corrosion resistance, and is widely used in aerospace industries [1–4]. The attractive combination of properties of 7055 alloy is attributed to high ratios of Zn/Mg, Cu/Mg and alloying element content, which in turn leads to the difficulty of processing the alloy. The commonly observed second phases in as-cast Al–Zn–Mg–Cu alloy are η ($MgZn_2$), T ($Al_2Mg_3Zn_3$) or T ($Al_{32}(Mg,Zn)_{49}$), M ($Mg(Zn_2AlCu)$), S (Al_2MgCu) and $A1_7Cu_2Fe$ [5–14]. The type and intrinsic character of second phase are decided by alloy composition, solidification manner and heat treatment conditions. Up to now, many different views on the stoichiometry of T phase still exist. Mondolfo [9] thought that the quaternary T phase in Al–Zn–Mg–Cu was formed by T (Al_6CuMg_4) phase in Al–Cu–Mg alloy and T ($Al_2Mg_3Zn_3$) phase in Al–Zn–Mg alloy, which had the same crystal structure and were mutually dissolvable. Modal and Mukhopadhyay [10] discovered η ($MgZn_2$), T ($Al_2Mg_3Zn_3$) and S (Al_2CuMg) in as-cast alloy having the composition of (wt%) 1.8–2.3Mg–2.0–2.6Cu–0.16Zr–0.05Fe–0.05Si. Li et al. [11] found that T phase ($Al_2Mg_3Zn_3$) could extensively dissolve copper, and had a formula of $Al_{40}Cu_{15}Mg_{25}Zn_{20}$. Robson [12] believed that the quaternary phase was formed by dissolving Zn in S (Al_2CuMg) in 7050 alloy. Chen et al. [13] detected the mixed constituents of T $(AlCuZn)_{49}Mg_{32}$, η $(AlCuZn)_{2-3}Mg$ and S (Al_2CuMg) in as-cast 7055 alloy. Fan [14] observed $Mg(ZnCuAl)_2$ and Al_7Cu_2Fe in as-cast alloy (wt%) Al–6.3Zn–2.36Mg–2.01Cu–0.12Zr.

Many constituents remain in the alloy after the subsequent homogenization and processing, owing to the proximity of composition to the limit of solid solubility in these alloys [2,11–17], which can deteriorate the age hardenability, aid crack initiation and propagation and cause variable properties. Chen et al [13] revealed that η phase dissolved completely and T and S phases remained in 7055 alloy after pretreatment at 450 °C for 35 h. A phase transformation of primary particle from $Mg(Zn,Cu,Al)_2$ phase to Al_2CuMg phase was found in Al–Zn–Mg–Cu alloy homogenized at 460 °C [14]. The high magnetic field of 12T could promote the dissolution of both T and S in Al–Zn–Mg–Cu alloy during homogenization [16,17]. Although the microstructures and properties of Al–Zn–Mg–Cu alloys have been investigated several decades, there still exist some problems about homogenization treatment. The nature of the constituent phase present in as-cast or annealing microstructure of Al–Zn–Mg–Cu alloy can provide a basis for the subsequent processing. In the present work, the evolutions of the conductivity, tensile properties and chemical composition of T phase during the subsequent homogenization of

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7050 alloy were studied in detail by FESEM, EDS, EPMA and XRD analysis.

2. Experimental

The composition of 7050 alloy used in this work is (wt%) Al–6.4Zn–2.22Mg–2.24Cu–0.11Zr–0.04Si–0.08Fe–0.10Mn–0.04Cr–0.05Ni–0.06Ti, and was produced by semi-continuous cast with the ingot size of 400 mm × 1400 mm. Five types of homogenization conditions were conducted: (1) 380 °C/2 h, (2) 400 °C/2 h, (3) 420 °C/2 h, (4) 440 °C/2 h and (5) 450 °C/2 h, and followed by air cooling. The heating rate is 5 °C/min. The temperature variation was controlled within ± 3 °C.

The conductivity of alloy at each condition was determined by the measurements of five specimens using a Fischer Sigmascope SMP10 type machine. Tensile specimens with a gauge diameter of 8 mm and a length of 30 mm were machined from the homogenized rod, and then tensile tested at room temperature using an Instron 1185 machine operating at a constant crosshead speed of 1.0 mm/min. The tensile properties at each condition were the average value of four specimens.

Samples for microstructure observations were prepared by standard metallographic methods and examined on a Zeiss Ultra Plus 60 type field-emission scanning electron microscopy (FESEM) having equipped with an Oxford Aztec 50 type energy dispersive X-ray analyzer having an image resolution of 0.8 nm. The area fraction of secondary phase in as-cast alloy and the alloy homogenized at different temperatures were measured by the average of 15 SEM photographs, and was calculated using Image J software. A quantitative X-ray wavelength dispersive spectroscopy (WDS) system attached to the SHIMADZU EPMA-1600 type instrument was used to analyze the chemistry of the secondary phases present in as-cast and homogenized microstructures. Phase identifications in as-cast and homogenized alloys were examined by X-ray diffraction analysis (XRD) with $CuK_{\alpha 1}$ radiation on PW3040/60X diffractometers.

3. Results and discussion

3.1. The evolutions of mechanical properties during homogenization

The conductivity and tensile properties of 7050 alloy changing with homogenizing temperature are displayed in Fig. 1. The conductivity (Fig. 1a) decreases by 22% when the as-cast alloy is homogenized at 380 °C/2 h, and then decreases gradually with increasing homogenized temperature, and finally reaches a stable value at 440 °C/2 h. While, σ_b, $\sigma_{0.2}$ and δ increase by 4.1%, 5.7% and 48.8%, respectively, after the as-cast alloy is homogenized at 380 °C/2 h, and then increases gradually with increasing temperature, and finally reaches the saturation values at 440 °C/2 h.

The tensile fracture surfaces of as-cast alloy and the alloys homogenized at 380 °C/2 h and 440 °C/2 h, respectively, are illustrated in Fig. 2. The secondary electron images (Fig. 2a, c and e) show a mixture of plenty of dimples with various sizes and intergranular cracks along grain and interdendritic grain boundaries on the tensile fracture surfaces of samples at all conditions, and the number of dimple increases with increasing temperature (Fig. 2c and e). Many gray secondary phases (circle region) can be clearly seen in the coarse dimples as shown in the backscattered electron images (Fig. 2b, d and f), and the size and amount of them decrease with increasing temperature. The EDS analysis indicates that these secondary phases contain elements Al, Zn. Mg and Cu or Al, Cu and Mg.

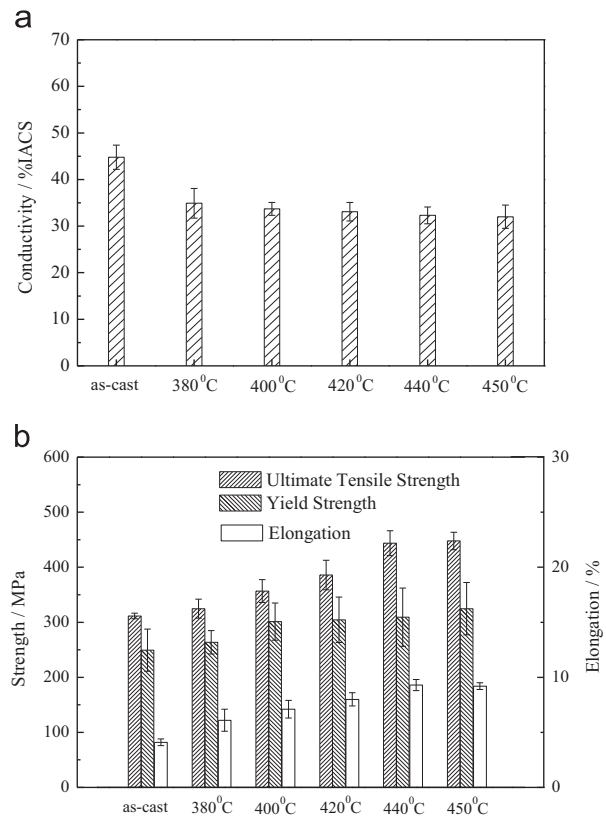

Fig. 1. The evolutions of conductivity (a) and tensile properties (b) with homogenized temperature.

3.2. The morphology of the quaternary phase T in as-cast alloy

The XRD analysis results of as-cast 7050 alloy are shown in Fig. 3. The XRD peaks (Fig. 3a) could be identified due to α (Al) solid solution (labeled as Al), $MgZn_2$ (labeled as η), Al_2CuMg (labeled as S) and Al_7Cu_2Fe. Fig. 3b illustrates the X-ray diffraction peaks of η phase, and the errors due to the drifting of the diffraction peaks of η phase are given in Table 1. It can be seen that the diffraction peaks of (100), (002) and (101) of η phase move to the lower angles. The deviation in the diffraction peaks of η phase between the standard and measured values is about 0.1°, and the error of the interplanar distance between the standard and measured values is about 0.02. The results indicate that the interatomic substitution happens in η phase [18]. It is reported that Cu and Al atoms can be dissolved in η phase and substitute for Zn atoms at the Zn sublattice to form a quaternary of $(CuZnAl)_2Mg$, which has the same crystal structure as $MgZn_2$ phase [14,15].

The typical as-cast microstructure of 7050 alloy consists of a cored dendritic structure of primary α(Al) surrounded by interdendritic secondary phases (Fig. 4a). Three types of secondary phases are observed in secondary electron image (Fig. 4b): irregular shape white phases A–C, irregular shape dark gray phase D, and strip gray phase E. The contrasts of these phases become brighter in backscattered electron image (Fig. 4c). The chemical compositions of secondary phases A–E obtained by EDS analysis are listed in Table 2. Because of the small size of the secondary phases, the electron beam is easy to detect the neighboring α (Al) matrix during the procedure of EDS analysis, and then the composition of matrix F is given as (wt%): 2.7 Mg, 87.5 Al, 4.4 Cu and 5.4 Zn. It can be seen that white phases A–C with the largest amount have the composition range of (wt%) 13–18.2 Mg, 25–41 Al, 17.4–25.1 Cu and 26–31.7 Zn. This confirms the fact that η phase dissolves considerable amount of Cu in it, which is the obvious

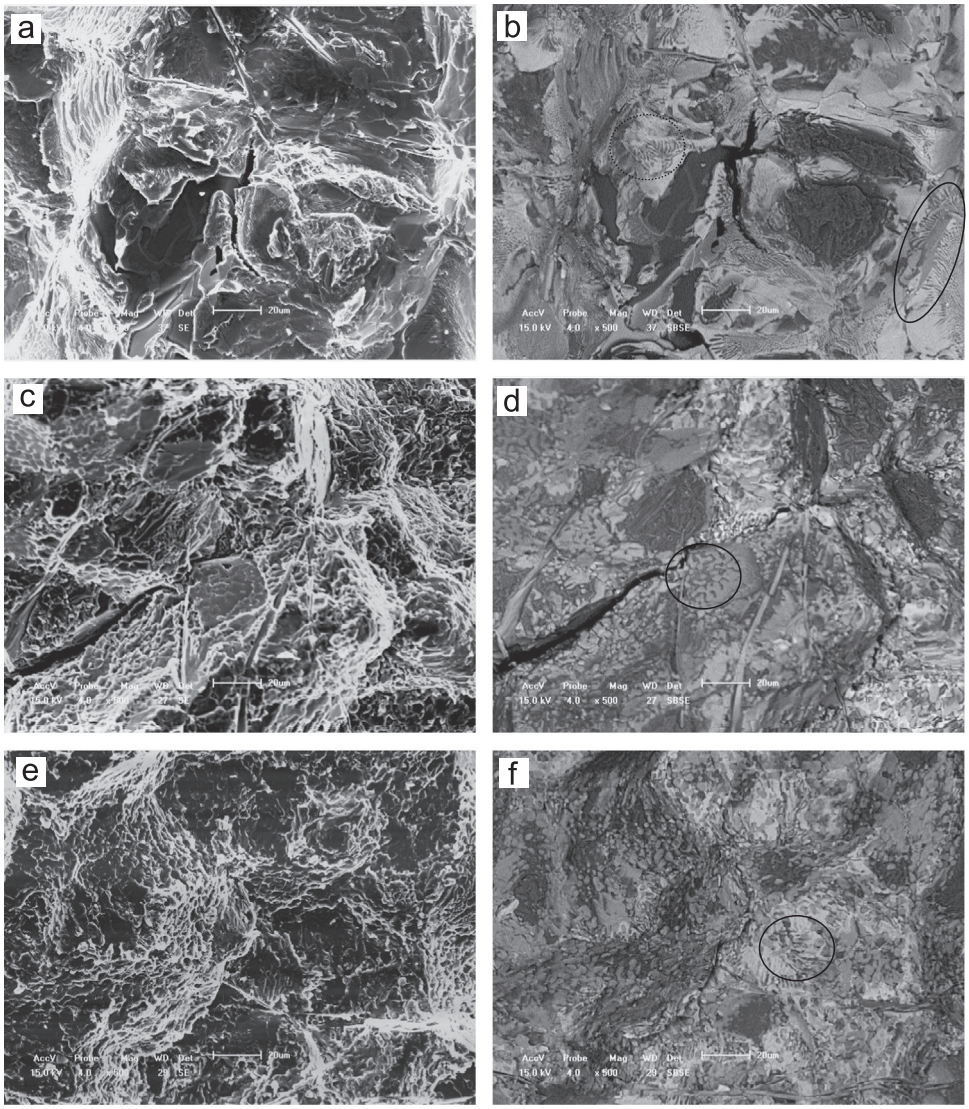

Fig. 2. Secondary electron images (a), (c) and (e) showing fracture mode and back-scattered images (b), (d) and (f) showing secondary phases on tensile fracture surface: (a) and (b) as-cast; (c) and (d) 380 °C/2 h; (e) and (f) 440 °C/2 h.

characteristics of the T phase. Dark gray phase D with quite small amount has a composition of (wt%) 11.6 Mg, 46.8 Al, 38.7 Cu and 2.9 Zn, and is determined to be S (Al_2CuMg), which indicates that S phase dissolves a small amount of Zn. Gray phase E with the least amount contains (wt%) 1.6 Mg, 54.4 Al, 23.5 Cu, 3.5 Zn and 17.1 Fe, and is identified to be Al_7Cu_2Fe.

The mole ratio of the quaternary phase T in as-cast 7050 alloy obtained by EPMA analysis is listed in Table 3, and it lies in the range of (mol%): 23.24–34.97 Mg, 28.10–48.92 Al, 11.80–17.09 Cu and 15.98–19.82 Zn. So, the stoichiometry of T phase in the present work can be determined as Al_4Mg_2ZnCu and $Al_3Mg_3Zn_2Cu_2$. It is quite difficult to find S phase using EPMA equipment because of its small size and amount.

The chemical compositions of T and S phases in Al–Zn–Mg–Cu alloy reported in literatures are listed in Table 4. The chemical composition of T phase in as-cast Al–Zn–Mg–Cu alloy normally falls in the range of (wt%) 13.8–30 Mg, 11.6–46 Al, 13.1–28.1 Cu and 23–41.4 Zn [10,13,15,20] or (at%) 19.5–37 Mg, 25–51.5 Al, 10.6–15.8 Cu and 16.4–22.2 Zn [8,21,23,24], which is consistent with the result in the present work. The chemical composition of T phase obtained in [8,10] is very close to that of the equilibrium T phase, having the same atomic fraction of Mg and Cu+Zn. It should be noted that the atomic fractions of Cu or Zn given in literatures [14,22,25] are much lower than those in the equilibrium T phase.

The composition of S phase obtained in the present work lies in the reported composition range of S phase (at%): 23.4–27.8 Mg, 47.1–57.4 Al, 17.3–21.1 Cu and 1.9–4 Zn [19,21]. It can be seen that the stoichiometry of S phase in as-cast Al–Zn–Mg–Cu approaches that of the equilibrium S phase. Nevertheless, the S phase reported in literature [10] has much higher Zn content, is still in the reported composition range of T phase [10,13,15,20].

3.3. The evolution of the morphology of T phase during homogenization

Fig. 5 demonstrates the XRD traces of 7050 alloy homogenized at 380 °C/2 h, 400 °C/2 h, 420 °C/2 h, 440 °C/2 h and 450 °C/2 h. The XRD peaks of α(Al), T, S, and Al_7Cu_2Fe are detected in the homogenized 7050 alloy. With increasing temperature, the amount and the height of diffraction peaks of T phase decrease, and while those of S phase increase.

Secondary and backscattered electron images (Fig. 6) display the changes in the morphology of T phase in the alloy homogenized at 380 °C/2 h. Three kinds of zones can be seen within

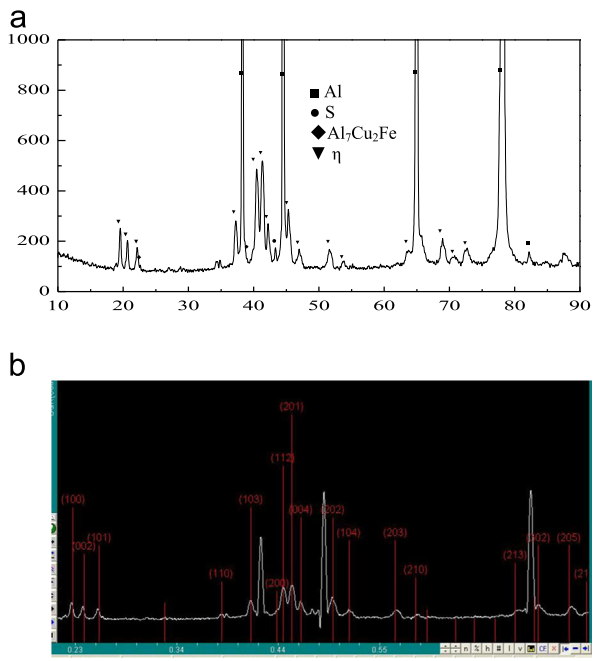

Fig. 3. X-ray diffraction results of (a) as-cast 7050 alloy; (b) the diffraction peaks of η phase.

Table 1
The errors due to the drifting of the diffraction peaks of η phase in Fig. 3(b).

Orientation index	2θ			Interplanar distance		
	Standard (deg)	Measured (deg)	Error	Standard (nm)	Measured (nm)	Error
(100)	19.660	19.565	0.095	4.5118	4.5335	0.0217
(002)	20.781	20.667	0.114	4.2709	4.2942	0.0233
(101)	22.263	22.130	0.133	3.9899	4.0135	0.0236

T phase: light gray zone (labeled as 1) situated at the periphery of T phase, bright white zone (labeled as 2) situated within T phase, and gray and dark gray zones (labeled as 3 and 4 respectively) situated at the margin of T phase. An elemental diffusion cell network (within zone 2) is clearly observed within T phase in the secondary electron image (Fig. 6a), which cannot be observed in as-cast microstructure (Fig. 4b). It should be noted that the secondary and backscattered electron images are good at observing the elemental diffusion cell network and secondary phases, respectively. The elemental diffusion cell network fades significantly within zones 3 and 4. The composition of zone 1 is (wt%) 2.3–5.8 Mg, 72.2–82 Al, 5.4–14.8 Cu and 6.3–7.2 Zn. The high

Table 2
Chemical compositions of secondary phases A–F (Fig. 4) in as-cast alloy obtained by EDS analysis (wt%).

	Mg	Al	Cu	Zn	Fe	Closest phase
A	14	40	20	26	–	T
B	18.2	25	25.1	31.7	–	T
C	13	41	17.4	28.6	–	T
D	11.6	46.8	19.7	2.9	–	S
E	1.6	54.4	23.5	3.5	17.1	Al_7Cu_2Fe
F	2.7	87.5	4.4	5.4	–	Matrix

Table 3
Chemical composition of the quaternary phase T obtained by EPMA analysis in as-cast alloy (mol%).

Phase present	Mg	Al	Cu	Zn
T	30.97	36.54	13.35	19.14
T	30.37	37.74	13.61	18.26
T	23.74	47.65	11.99	16.45
T	23.24	48.92	11.8	15.98
T	34.96	28.1	17.09	19.82
T	33.49	30.69	16.26	19.56

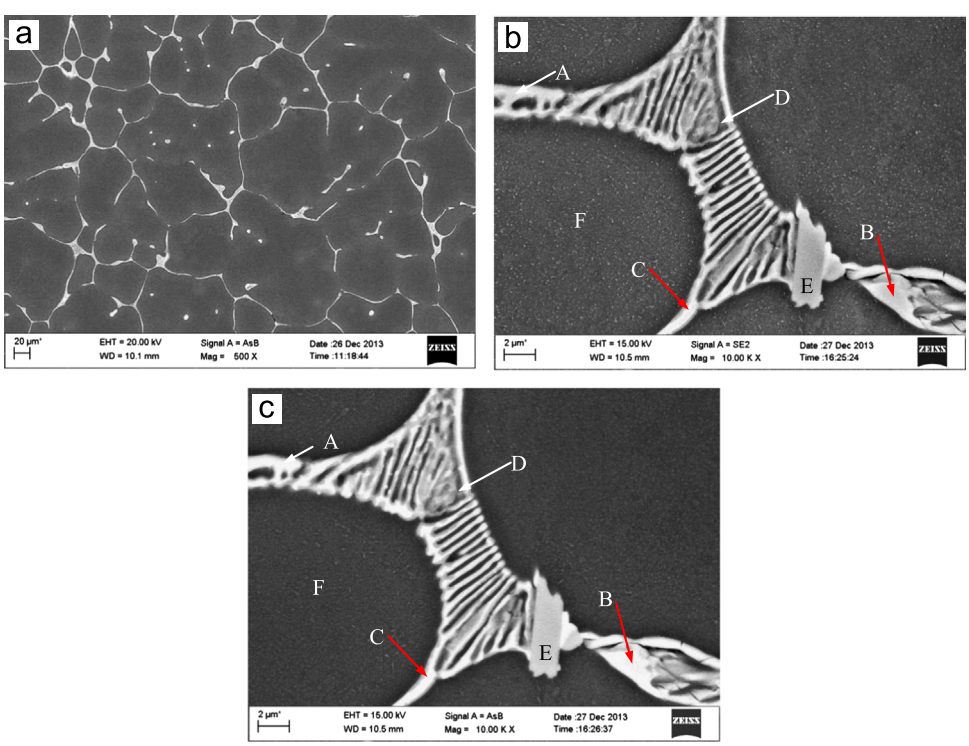

Fig. 4. Microstructure of as-cast 7050 alloy: (a) the dendritic microstructure; (b) and (c) secondary backscattered electron images of interdendritic secondary phases at high magnification.

Table 4
Chemical compositions of secondary phases in Al–Zn–Mg–Cu alloy reported in literature.

Alloy composition	State	Phase	Mg	Al	Cu	Zn
Al–5.1Zn–2.5Mg–1.2Cu [15]	Cast	T (wt%)	13.8–18.1	31.5–46	15.6–19.9	24.5–30.5
7055 [8]	Cast	T (at%)	37	25	15.8	22.2
7050 [19]	Cast	S (at%)	24.2–27.8	47.1–53.5	18.8–21.1	3.5–4
	400 °C	S (at%)	19.1–28	48.9–61.6	18.2–18.6	0.6–4.7
	Stepped annealing	S (at%)	23.1–24.4	51.6–54.9	21.7–23.8	0.2–0.26
AA7075 [20]	Cast	T (wt%)	14.8	41.5	13.1	30
AA7085 [21]	Cast	T (at%)	27	39	15	19
		S (at%)	23.4	57.4	17.3	1.9
Al–8.1Zn–2.1Mg–2.3Cu [22]	Cast	T (wt%)	16.6–20.2	51.3–62.4	8–10.4	13–17.7
7B04 [23]	Cast	T (mol%)	21.7	47.7	10.6	20
	470 °C	T (mol%)	32.4	26.7	20.4	20.5
		S (mol%)	21.3	58.1	20.6	–
7055 [24]	Cast	T (at%)	19.5	51.5	12.8	16.4
	460 °C	S (at%)	15.3	57.7	19.7	7.3
7055 [14]	Cast	T (mol%)	16.9–19.4	51.5–68.3	6.6–12.8	8–16.4
	460 °C	S (mol%)	21.3	45.7	29.9	3.1
Al–10Zn–2.5Mg–2.5Cu [25]	Cast	T (at%)	17–20	51–57	5–9	15–20
	460 °C	T (at%)	33.7–34.4	14.2–14.7	10.5–14.9	34–41.3
7055 [10]	Cast	T (wt%)	20.1–23.1	11.6–17.4	23.2–28.1	35.3–41.4
		S (wt%)	12.9–15.2	37–49.6	17.2–23.1	24.6–30.7
	450 °C	T (wt%)	19.1–21.1	15.3–20.2	24.8–26	33.5–37.6
		S (wt%)	16–17.1	40.3–47	34.1–39.7	2.7–3.2
Al–8Zn–7Mg–2Cu [6]	400 °C	T (wt%)	22.1–23.9	25.5–27.2	13.1–17.6	34.9–36.8
		T (at%)	34–35.9	35.4–36.9	7.5–9.4	20–20.9
7050 [12]	400 °C	S (at%)	18.2	59.3	20.4	2.1
7055 [13]	Cast	T (wt%)	30	33	14	23
	450 °C	T (wt%)	35	26	16	23

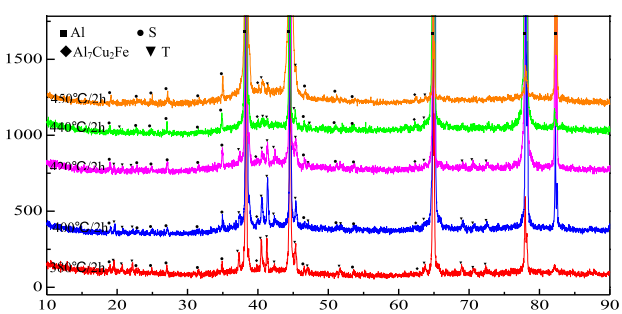

Fig. 5. XRD traces of 7050 alloy homogenized at 380 °C, 400 °C, 420 °C, 440 °C and 450 °C for 2 h.

magnification image (Fig. 6c) reveals that the cell network consists of white cell wall A and light gray cell interior B. The composition range of cell wall A is (wt%) 12–19 Mg, 18–34 Al, 22–28 Cu and 30–42 Zn. Cell interior B has a composition range of (wt%) 10–18 Mg, 24–32 Al, 23–30 Cu and 17–23 Zn. Comparing with the composition of T phase in as-cast 7050 alloy, besides having same Mg and Cu contents, cell wall A has higher Zn and while cell interior B has lower Zn. Zone 3 has a composition range of (wt%) 12.2–14 Mg, 44.9–51.6 Al, 20.4–27.4 Cu and 11.4–15.8 Zn. Zone 4 with deeper contrast contains (wt%) 13.8–15.2 Mg, 43.9–48.1 Al, 26–33.3 Cu and 5.6–10 Zn, and has lower Zn and same Mg and Cu than those in zone 3.

Fig. 7 demonstrates the evolution of the morphology of T phase in 7050 alloy homogenized at 400 °C/2 h, 420 °C/2 h, 440 °C/2 h and 450 °C/2 h, respectively. With increasing temperature, the cell network weakens gradually within bright white, gray and dark gray phases when the temperature is $\leq$420 °C (Figs. 6a and 7a and c) and nearly disappears within dark gray phase when the temperature is $\geq$440 °C (Fig. 7e and g); the amount and the size of gray and dark gray phases increase and the difference in contrasts between gray and dark gray phases decreases (Fig. 7b, d, f and h). The changes of the compositions of the phases within T phase (according to the contrast: bright white, gray and dark gray) with the homogenization temperature are given in Table 5. The composition of the bright white phase at different temperatures changes little, and is the same as that of T phase in as-cast 7050 alloy. With increasing temperature, Cu content increases and Zn content decreases gradually in the gray and dark gray phases, and the deviation in the compositions of the two phases reduces. The composition of phase 16 is (wt%) 14.6 Mg, 35.7 Al, 45.9 Cu and 3.8 Zn when the temperature is 450 °C, and is close to that of the equilibrium S phase. Because of the dissolution and the transformation of T phase, many needle Al_7Cu_2Fe phases are revealed (Fig. 7e and f), which always precipitate along with T phase during the solidification of ingot, and so it is difficult to be detected in the microstructures of alloy homogenized lower than 420 °C.

The compositions of T and S phases in homogenized Al–Zn–Mg–Cu alloy reported in literatures are given in Table 4. The T phase generally falls: (wt%) 19.1–35 Mg, 15.3–27.2 Al, 13.1–26 Cu and 23–37.6 Zn [6,10,13] or (at%) 32.4–35.9 Mg, 14.2–36.9 Al, 7.5–20.4 Cu and 20–41.3 Zn [6,23,25]. The composition of T phase in the present work is consistent with that reported by Mondal et al. [10]. The S phase normally falls: (wt%) 16–17.1 Mg, 40.3–47 Al, 34.1–39.7 Cu and 2.7–3.2 Zn [6] or (at%) 18.2–28 Mg, 45.7–61.6 Al, 18.2–29.9 Cu and 0–7.3 Zn [12,14,19,23,24], which is close to the compositions of gray or dark gray phases in the present work.

During homogenization, the region within T phase such as zone 1 which has much lower Cu, Mg and Zn contents sluggishly dissolves into the matrix. At the same time, the elemental diffusion cell network begins to appear within T phase, and consists of the higher Zn white cell wall and the lower Zn light gray cell interior, with the same Mg and Cu contents as those of T phase. The zone having Zn content higher than 10 wt% is called T-base phase in the present work. The appearance of the elemental diffusion cell network demonstrates that Zn in the cell interior is diffusing out to the cell wall, and the cell interior is at the transitive stage of the phase transformation from T to S. Many authors investigated the microstructural evolution of Al–Zn–Mg–Cu alloy during homogenization [6,10,12–14,19,23–25], but they

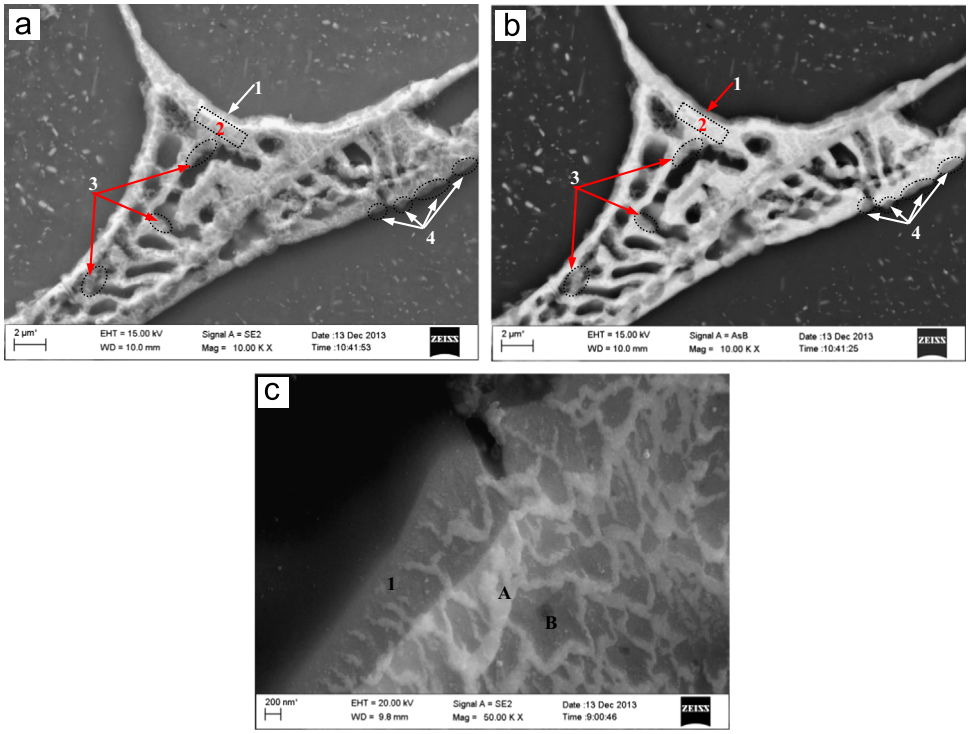

Fig.6. FESEM micrographs of 7050 alloy homogenized at 380 °C/2 h showing the morphology of T phase and (b) secondary and backscattered electron images, (c) high magnification image of network.

did not report the existence of the elemental diffusion cell network within T phase. With the progress of homogenization, Zn content within T phase continues to reduce, the light gray cell interior changes to the gray phase and then to the dark gray phase. The network fades when the transformation from the cell interior to the gray phase happens, and disappears when the transformation from the gray phase to the dark gray phase occurs. With increasing temperature, the amount and the size of the gray and dark gray phases increase; the difference in the composition between the two phases reduces. The composition of the dark gray phase finally approaches the equilibrium S phase at 450 °C. The gray phase contains Zn higher than 10 wt%, and is still T-based phase when the temperature is $\leq$420 °C, and changes to S-based phase when the temperature is $\geq$440 °C. The existence of the elemental diffusion cell network indicates the progress of phase transformation from T to S, and it disappears when the phase transformation finishes. It should be noted that the diffusion of Zn happens even at 380 °C, and while the obvious diffusions of Mg and Cu begin at 440 °C. Du et al. summarized and evaluated the measured impurity diffusions of Cr, Mn, Fe, Ni, Cu, Zn, Mg and Si in fcc Al [26]. The normally used diffusion coefficients (D_0) and activation energy (Q) for Zn, Mg and Cu are listed in Table 6. It can be seen that the diffusion activation energy of Zn is close to that of Mg, and much lower than that of Cu, and while Mg has the smallest D_0 value among them. Therefore, we can determine that the diffusion velocity of Zn is higher than those of Mg and Cu, and the transformation from T to S is mainly controlled by the diffusion of Cu.

3.4. The evolutions of the area fractions of T and S during homogenization

Since the amount of Al_7Cu_2Fe phase is extremely small, and it changes little during homogenization because of its high melting temperature, thus the area fraction of Al_7Cu_2Fe phase can be ignored in the measurement of the area fraction of the secondary phases. The evolutions of the area fraction of the secondary phase in as-cast and homogenized alloys are given in Table 7. The area fraction of the secondary phase in as-cast alloy is 7.39%. Because of the small amount of S phase in as-cast alloy, it is reasonable to consider that the area fraction of the secondary phase in as-cast alloy is T phase. The area fraction of the secondary phase suddenly reduces to 7.17% at 380 °C/2 h, and then gradually to 5.15% at 450 °C/2 h. The area fraction of T phase decreases rapidly from 5.49% at 380 °C/2 h to 2.18% at 440 °C/2 h and then gradually to 1.27% at 450 °C/2 h, and while, that of S phase increases significantly at 440 °C/2 h. The value of S/T (the ratio between the area fractions of S and T) is used to evaluate the degree of the phase transformation.

Profuse coarse secondary phase T and small amount of S and Al_7Cu_2Fe precipitated as eutectic structures with α(Al) matrix form a network at grain and interdendritic grain boundaries in the as-cast alloy. According to the tensile fracture surface observations, many coarse secondary phases are located in large dimples in the as-cast alloy; it is evident from Fig. 4a and b that the interface between the coarse secondary phase and the matrix is ripped first and acts as premature microcrack, and then links up along grain boundaries which has been reported previously [30]. The fracture mode of as-cast alloy is mainly intergranular, and the strengths and elongation of as-cast alloy are low.

During homogenization, some regions within T phase having much lower contents of Zn, Mg and Cu sluggishly dissolve into the nearby matrix, some regions within T phase gradually transform to S phase and this phase transformation becomes significant when the temperature is $\geq$440 °C. With increasing homogenization temperature, T phase becomes discontinuous and grows into isolated single phase particle, the size and amount of T phase decrease, while the amount of S phase increases and obvious coarsening of S phase starts at 440 °C. The dissolution of the coarse secondary phases by homogenization permits the matrix to deform uniformly, avoiding early fracture and producing plenty of fine dimples; the transgranular fracture becomes predominant

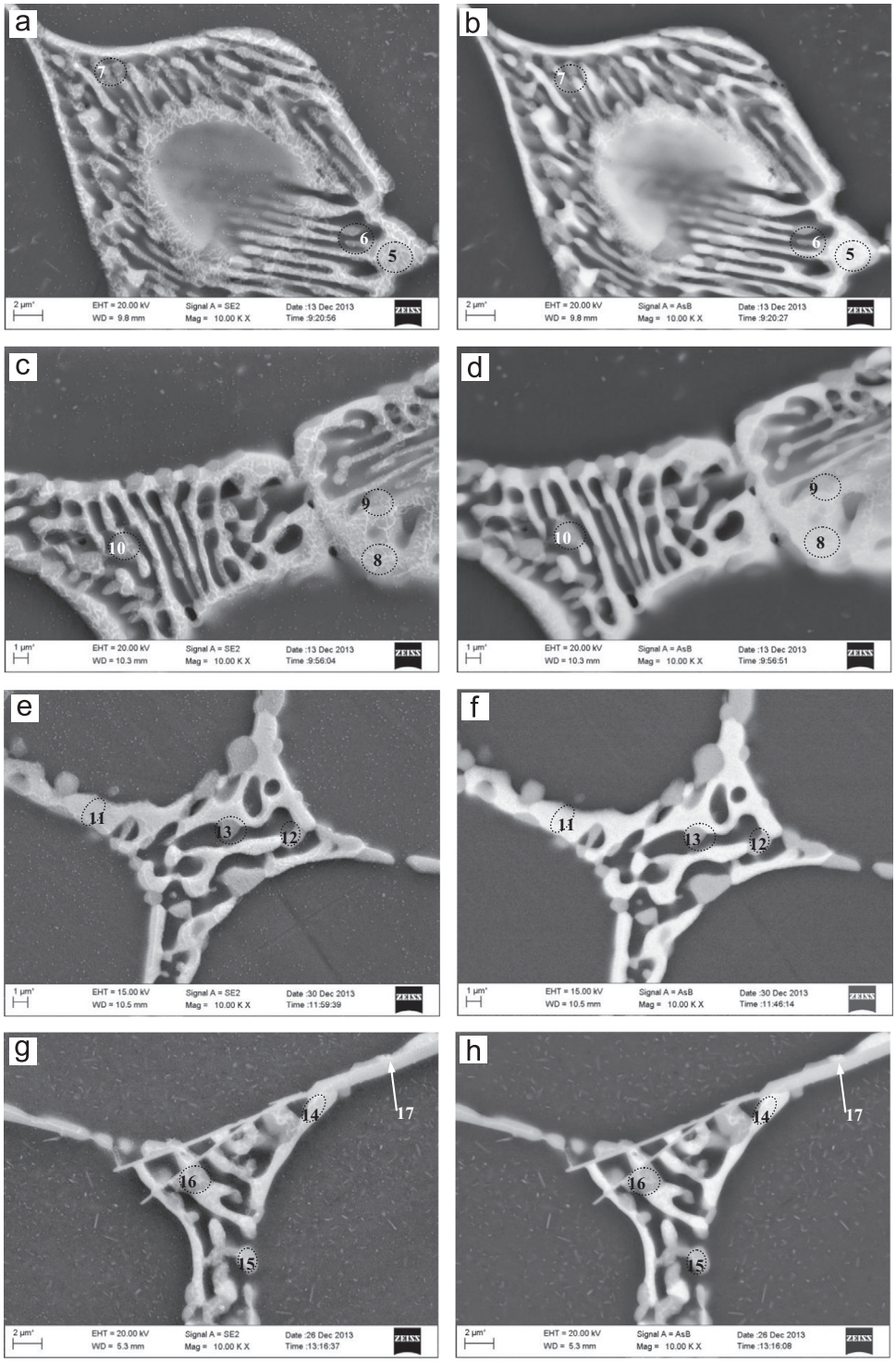

Fig. 7. FESEM secondary and backscattered electron images showing the changes of the morphologies of T phase in 7050 alloy homogenized at (a) and (b) 400 °C/2 h; (c) and (d) 420 °C/2 h; (e) and (f) 440 °C/2 h; (g) and (h) 450 °C/2 h.

in homogenized alloy, and thus the elongation and yield strength are greatly improved. Furthermore, in progress of the dissolution of secondary phases, more and more alloying elements Zn, Mg and Cu come into the α(Al) matrix, and so the strengths increase with increasing homogenization temperature as a result of solid solution hardening [31]. According to the Mathiessen theory [9], the electrical conductivity of the alloy can be expressed as follows: $\rho=\rho_0+\Delta\rho_s+\Delta\rho_p+\Delta\rho_v+\Delta\rho_d+\Delta\rho_g$, where, ρ_0, $\Delta\rho_s$ and $\Delta\rho_p$ are the electrical conductivity caused by Al matrix, solid soluble atoms and precipitates, respectively. $\Delta\rho_v$, $\Delta\rho_d$ and $\Delta\rho_g$ are the electrical conductivity caused by vacancies, dislocations and grain boundaries, respectively. The contribution of $\Delta\rho_s$ to the electrical conductivity is the greatest among them [32]. The ability of scattering electron increases when the amount of solute atoms dissolved in Al matrix increases, and so the electrical conductivity decreases with increasing homogenization temperature.

It should be noted that the conventional homogenization temperature of 7050 alloy is between 450 °C and 470 °C; S phase accounts for more than 77% in the secondary phase according to our results, which is realized by few researchers. The results of the

Table 5
Chemical compositions of secondary phases in 7050 alloy at different homogenized conditions (wt%).

		Mg	Al	Cu	Zn	Fe	Phase contrast	Closest phase
380 °C/2 h	2	15.2	27.8	24.7	32.3	–	Bright white	T
	3	13.0	45.7	26.6	14.7	–	Gray	T-base
	4	14.3	48.1	28.1	9.5	–	Dark gray	S-base
400 °C/2 h	5	15.2	24.8	26.9	33	–	Bright white	T
	6	11.1	47.9	28.5	12.5	–	Gray	T-base
	7	12.6	46	32.1	9.1	–	Dark gray	S-base
420 °C/2 h	8	15.2	27.8	24.7	32.3	–	Bright white	T
	9	10.1	50.5	28.9	10.5	–	Gray	T-base
	10	10.9	46.9	34.7	7.5	–	Dark gray	S-base
440 °C/2 h	11	14.2	27.9	25.4	32.5	–	Bright white	T
	12	14.5	36.1	40.8	8.6	–	Gray	S-base
	13	11.2	44.4	37.7	6.7	–	Dark gray	S-base
450 °C/2 h	14	15.3	25.6	23.3	35.8	–	Bright white	T
	15	12.1	42.8	38.9	6.2	–	Gray	S-base
	16	14.6	35.7	45.9	3.8	–	Dark gray	S-base
	17	0	63.9	25.3	2.6	8.19	Bright gray	Al_7Cu_2Fe

Table 6
The diffusion coefficients (D_0) and activation energy (Q) for Zn, Mg and Cu.

	D_0 (m^2/s)	Q
Zn [27]	2.6×10^{-5}	119.1
Mg [28]	6.2×10^{-6}	114.7
Cu [29]	6.5×10^{-5}	136.1

Table 7
The area fraction of secondary phase in as-cast and homogenized 7050 alloys.

	As-cast	380 °C/2 h	400 °C/2 h	420 °C/2 h	440 °C/2 h	450 °C/2 h
Residual phase	7.39%	7.17%	6.66%	5.73%	5.48%	5.15%
T phase	7.39%	5.49%	4.85%	3.8%	2.18%	1.27%
S phase	0	1.68%	1.81%	1.93%	3.3%	3.88%
S/T	0	0.31	0.37	0.51	1.51	3.06

present work will provide useful information for material researchers to determine the proper homogenization condition for Al–Zn–Mg–Cu alloy.

4. Conclusions

(1) The conductivity decreases by 22%, while σ_b, $\sigma_{0.2}$ and δ increase by 4.1%, 5.7% and 48.8%, respectively, when the as-cast alloy is homogenized at 380 °C/2 h, and finally reaches the saturation values at 440 °C/2 h.
(2) The as-cast microstructure of 7050 alloys contains eutectic α (Al)+T (Al_4Mg_2CuZn or $Al_3Mg_3CuZn_2$), S (Al_2CuMg) and Al_7Cu_2Fe. The deviation in the diffraction peaks of η ($MgZn_2$) phase between the standard and measured values is about 0.1°, which confirms that Cu and Al atoms substitute for Zn atoms at the Zn sublattice to form a quaternary phase T.
(3) During homogenization, the region within T phase having much lower Cu, Mg and Zn contents dissolves into the matrix. At the same time, the elemental diffusion cell network appears due to the diffusion of Zn from the light gray cell interior to the white cell wall, with the same Mg and Cu contents as those of T phase. With increasing temperature, the network fades when the transformation from the light gray cell interior to gray phase and then to the dark gray phase occurs, and finally disappears when the composition of the dark gray phase approaches the equilibrium S phase; and the amount and the size of the gray and dark gray phases increase, and the difference in composition between the gray and dark gray phases reduces. The diffusion of Zn happens even at 380 °C, while the obvious diffusions of Mg and Cu begin at 440 °C. The above results indicate that the diffusion velocity of Zn is higher than that of Mg and Cu, and the transformation from T phase to S phase is mainly controlled by the diffusion of Cu.
(4) The area fraction of T phase decreases rapidly from 5.49% at 380 °C/2 h to 2.18% at 440 °C/2 h, and then slowly to 1.27% at 450 °C/2 h, and while that of S phase increases significantly at 440 °C/2 h.

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